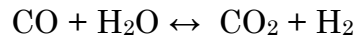


(1) At low to moderate pressures, the equilibrium state of the water-gas shift reaction



Is approximately described by the relation

$$\frac{x_{\text{CO}_2} x_{\text{H}_2}}{x_{\text{CO}} x_{\text{H}_2\text{O}}} = K_e(T) = a \exp\{b/T\} ; a = 0.0247 \text{ and } b = 4020 \text{ K}$$

where T is the reactor temperature in Kelvin, K_e is the reaction equilibrium constant, and x_i are the mole fractions of species i in the reactor at equilibrium. The feed to the reactor contains 20 mole % CO, 10 mole % CO₂, 40 mole % water, and the balance is inert gas. The reactor is maintained at $T = 1123 \text{ K}$.

- Assume a basis of 1 mole of feed and draw and label a flowchart. Carry out a degree-of-freedom analysis of the reactor based on the extent of reaction.
- Calculate the total moles of gas in the reactor at equilibrium and then the equilibrium mole fraction of hydrogen in the product.
- Suppose a gas sample is drawn from the reactor and analyzed shortly after startup and the mole fraction of hydrogen is significantly different from the calculated value. Assuming the calculations are correct, what is a likely explanation for the discrepancy?
- Use a numerical method of your choice to take the input of the reactor temperature and the input feed component mole fractions of CO, H₂O, and CO₂ (assume no hydrogen is fed and no inert is present) and to calculate the mole fraction x_{H_2} in the product gas when equilibrium is reached. Present your results in a table. Use your numerical method to try and maximize the yield of hydrogen by adjusting the temperature and feed composition.

(2) Murphy 3.51 (Use reaction conversion information from Murphy 2.69)

(3) Murphy 4.10

(4) Murphy 4.19